04 · Case 3, part 2: keeping dust out, and bringing the pressure down

The words of this track, as it says them. Bold brown is the examiner.

Exam case 3, continued. Ane's airway defences, and Ane's blood pressure.

Three marks. How does the body usually keep dust and small particles out of the lungs? There are several mechanisms.

The hook: stick and sweep, filter, cough.

Mucus and cilia: the muco-ciliary escalator. Nasal hairs: a filter that traps particles, which end up stuck in mucus. And the carina, at the bottom of the trachea, which sets off the cough reflex. The epiglottis keeps large objects out of the trachea, but it's no help against small particles.

Two marks. The muco-ciliary escalator is one of the mechanisms. How does it work?

Cilia, tiny hair-like structures lining the respiratory epithelium, beat to sweep mucus up and out of the airways. And mucus is sticky: dust and particles get stuck in it and trapped, and the cilia carry them up and out with it.

Her formative test gives it a tag line: the mechanism formed by the pseudostratified columnar epithelium lining the airways is the muco-ciliary escalator.

Two marks. Smoking damages this escalator. How would this affect a smoker?

The cilia are damaged, so mucus can't be moved up and out of the lungs. And a smoker's cough develops, to replace the lost function of the cilia.

Four marks. Ane's blood pressure was high because of the fight or flight response. What process helps bring it back to normal after the attack? Think the full process.

This is the baroreceptor reflex. Her recorded session talks it through, and she says think the full process, so give every link.

The hook: stretch, medulla, brake on, accelerator off, vagus slows the heart, vessels widen.

One. The carotid and aortic baroreceptors, which are stretch receptors in the blood vessel walls, pick up that the blood pressure is high. The walls are very stretched.

Two. They send the message to the medulla oblongata, to its cardio-acceleratory, cardio-inhibitory and vasomotor centres.

Three. The cardio-inhibitory centre is stimulated, and the cardio-acceleratory centre is inhibited. As she puts it, you can't brake and accelerate at the same time.

Four. The message goes down the vagus nerve. The rate and the force of the heartbeat drop, so the cardiac output drops.

Five. The vasomotor centre brings about vasodilation: more space for the blood.

And six: together, that brings the blood pressure back down to normal.

Once more. Ane's defences, and Ane's blood pressure.

The defences: mucus and cilia, the muco-ciliary escalator; nasal hairs that filter; the carina, which triggers the cough reflex; and the epiglottis, for large objects. The escalator: cilia on the respiratory epithelium beat to sweep sticky, dust-trapping mucus up and out. Smoking damages the cilia, so mucus can't be moved, and a smoker's cough replaces them. Blood pressure: the carotid and aortic baroreceptors sense the stretch and tell the medulla oblongata. The cardio-inhibitory centre is stimulated and the cardio-acceleratory centre inhibited, and the vagus nerve slows the rate and force, so cardiac output drops. The vasomotor centre causes vasodilation, and the pressure returns to normal.